

Coordinate-free Mathematics

Paul Gustafson

May 2025

The first section develops algebraic constructions independent of a basis; the second develops geometric constructions independent of coordinates; the third develops analytic constructions from norms, operators, and positivity.

1 Algebra

Linear algebra studies linear maps between vector spaces. A **vector space** V over a field k is a set equipped with vector addition and scalar multiplication operations satisfying straightforward compatibility conditions. A **linear map** $f : V \rightarrow U$ between k -vector spaces is a map of underlying sets such that $f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$ for all $\alpha, \beta \in k$ and $v, w \in V$.

The **free k -vector space on a set** X , denoted kX , is the set of finitely supported set-theoretical functions $X \rightarrow k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W , denoted V/W , is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is the vector space $k(V \times W)/N$ where N is the subspace generated by the obvious linearity relations on V and W .

An **isomorphism** of vector spaces $V \cong W$ consists of linear maps $f : V \rightarrow W$ and $g : W \rightarrow V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$.

Exercise 1 (Universal constructions)

Prove the following:

- $k(X \sqcup Y) \cong kX \oplus kY$
- $k(X \times Y) \cong kX \otimes kY$
- *The first isomorphism theorem for vector spaces (and second and third if you want).*
- *Find universal properties determining up to canonical isomorphism the free k -vector space, direct sum, quotient, tensor product, and dual space (defined below), respectively.*

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k . The evaluation map $\text{ev}_V : V^* \otimes V \rightarrow k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\text{coev}_V : k \rightarrow V \otimes V^*$ satisfying the snake equations $(1_V \otimes \text{ev}_V) \circ (\text{coev}_V \otimes 1_V) = 1_V$ and $(\text{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \text{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \text{tr}(1_V)$, where $\text{tr}(f) : \text{End}(V) \rightarrow k$ is given by $f \mapsto (\text{ev}_V \circ s_{V, V^*} \circ (f \otimes 1_{V^*}) \circ \text{coev}_V)(1)$ where $s_{X, Y} : X \otimes Y \rightarrow Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$.

Exercise 2 (Trace and dimension identities)

Prove the following for finite-dimensional vector spaces V, W and linear maps $f, g : V \rightarrow V$.

- $\text{tr}(fg) = \text{tr}(gf)$
- $\text{tr}(f \oplus g) = \text{tr}(f) + \text{tr}(g)$
- $\text{tr}(f \otimes g) = \text{tr}(f) \text{tr}(g)$
- $\dim(V) \in \mathbb{N}$
- $V \cong W \iff \dim(V) = \dim(W)$

Define the tensor algebra and **exterior algebra of V** by

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \quad \Lambda V = T(V) / \langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree- k component of ΛV .

Exercise 3 (The top exterior power is one-dimensional) Show that the highest graded component of ΛV is one-dimensional.

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)})$.

For an n -dimensional real vector space V , its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive.

Exercise 4 (Determinant identities and volume)

Prove the following:

- $\det(f \oplus g) = \det(f) \det(g)$
- If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0, 1]$.
- Cramer's rule

For $f \in \text{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ .

Exercise 5 (Characteristic polynomial and Jordan form)

Let $f \in \text{End}(V)$ with V finite dimensional. Suppose its characteristic polynomial splits over k . Prove:

- Every eigenvalue of f is a root of its characteristic polynomial.
- The geometric multiplicity of each eigenvalue is at most equal to its **algebraic multiplicity** (i.e. multiplicity as a root of the characteristic polynomial)
- $\text{tr}(f) = \sum_i \lambda_i$ and $\det(f) = \prod_i \lambda_i$, where each eigenvalue λ_i is repeated according to its algebraic multiplicity
- The operator f satisfies its own characteristic polynomial (replacing λ with f)
- The existence of the Jordan normal form for a linear operator on a finite-dimensional complex vector space

A **Hermitian inner product** on a complex vector space V is a positive-definite sesquilinear form $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable and linear in the second. It induces the norm $\|v\| = \sqrt{\langle v, v \rangle}$. For a linear map $A : V \rightarrow W$ between finite-dimensional Hermitian spaces, an **adjoint** is a map $A^* : W \rightarrow V$ satisfying

$$\langle Av, w \rangle = \langle v, A^*w \rangle.$$

An endomorphism A is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle v, Av \rangle \geq 0$ for every v , and **unitary** if $A^*A = AA^* = 1$. Equip $\text{End}(V)$ with the induced operator norm.

Exercise 6 (Finite-dimensional spectral theorem)

Prove that every linear map between finite-dimensional Hermitian spaces has a unique adjoint and establish the identities $(AB)^* = B^*A^*$ and $\|A^*A\| = \|A\|^2$. Prove that $A \in \text{End}(V)$ is positive if and only if $A = B^*B$ for some B , and prove that every normal A has an orthogonal spectral decomposition

$$A = \sum_{\lambda} \lambda p_{\lambda}, \quad p_{\lambda} p_{\mu} = \delta_{\lambda\mu} p_{\lambda}, \quad \sum_{\lambda} p_{\lambda} = 1.$$

Apply the theorem to T^*T to prove that every map $T : H \rightarrow K$ between finite-dimensional Hermitian spaces has a singular-value decomposition

$$T = U \Sigma V^*.$$

Prove that the diagonal entries of Σ are the square roots of the nonzero eigenvalues of T^*T , counted with multiplicity.

1.1 Group representations

An action of a group G on a set X assigns to each $g \in G$ a bijection $x \mapsto gx$ such that $1x = x$ and $(gh)x = g(hx)$. A **representation** of G on a vector space V is a homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

A subspace $W \subseteq V$ is **G -invariant** if $\rho(g)W \subseteq W$ for every $g \in G$. Restriction gives the **subrepresentation** W , and $\rho(g)(v + W) = \rho(g)v + W$ gives the **quotient representation** V/W . A nonzero representation is **irreducible** if its only invariant subspaces are 0 and V .

For representations V and W , the direct sum, tensor product, and dual representations are defined by

$$\begin{aligned} g(v, w) &= (gv, gw), \\ g(v \otimes w) &= gv \otimes gw, \\ (g\phi)(v) &= \phi(g^{-1}v), \quad \phi \in V^*. \end{aligned}$$

An **intertwiner** $T : V \rightarrow W$ is a linear map satisfying $T\rho_V(g) = \rho_W(g)T$ for every g . Write $\mathrm{Hom}_G(V, W)$ for the space of intertwiners and

$$V^G = \{v \in V : \rho(g)v = v \text{ for every } g \in G\}$$

for the fixed-point space.

Exercise 7 (Averaging produces invariant projections and inner products)

Let G be finite and let V be a complex representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

Prove that P_G is the projection of V onto V^G . Prove also that averaging any Hermitian inner product over G produces a G -invariant Hermitian inner product, so every finite-dimensional complex representation of G is equivalent to a unitary representation.

1.2 Maschke's theorem and Schur's lemma

Exercise 8 (Maschke's theorem)

Let $W \subseteq V$ be G -invariant and let $P : V \rightarrow W$ be any linear projection. Define

$$\tilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g)P\rho(g)^{-1}.$$

Prove that $\tilde{P}$ is G -equivariant and has image W . Deduce that $\ker \tilde{P}$ is a G -invariant complement of W . If G is finite and the characteristic of k does not divide $|G|$, prove that every finite-dimensional k -representation of G is a direct sum of irreducible representations.

Exercise 9 (Schur's lemma)

Let V and W be irreducible finite-dimensional complex representations. Prove that every nonzero intertwiner $T : V \rightarrow W$ is an isomorphism and that

$$\text{End}_G(V) = \mathbb{C}1_V.$$

Let $\mathbb{C}[G]$ act on V by the linear extension of ρ . Deduce that every element of the center $Z(\mathbb{C}[G])$ acts by a scalar on each irreducible representation.

1.3 Characters and orthogonality

The **character** of a finite-dimensional representation V is the function

$$\chi_V(g) = \text{tr}(\rho_V(g)).$$

For class functions $\chi, \psi : G \rightarrow \mathbb{C}$, define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 10 (Great Orthogonality Theorem)

Prove that characters are class functions and that, for unitary representations,

$$\chi_{V \oplus W} = \chi_V + \chi_W, \quad \chi_{V \otimes W} = \chi_V \chi_W, \quad \chi_{V^*}(g) = \overline{\chi_V(g)}.$$

Let Γ^α and Γ^β be irreducible unitary representations of a finite group G , of dimensions d_α and d_β . In orthonormal bases, prove that

$$\sum_{g \in G} \Gamma^\alpha(g)_{ij} \overline{\Gamma^\beta(g)_{kl}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl}.$$

Deduce that their characters satisfy

$$\langle \chi_\alpha, \chi_\beta \rangle_G = \delta_{\alpha\beta}.$$

If V is a finite-dimensional representation, prove that the multiplicity of V_α in V is

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G.$$

Conclude that the character determines a finite-dimensional complex representation up to isomorphism.

For an irreducible character χ_α , define the **symmetry projection**

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g).$$

Exercise 11 (Symmetry projections)

Prove that the operators P_α are pairwise orthogonal idempotents and that $P_\alpha V$ is the α -isotypic component of V . Interpret P_α as constructing symmetry-adapted linear combinations of atomic orbitals or displacement vectors.

1.4 Molecular symmetry

The **point group** of a molecule is the finite group $G \subseteq O(3)$ of spatial symmetries preserving its nuclear configuration, including atomic species. For a molecule with N atoms, G acts on the real displacement space

$$V_{\text{disp}} = \bigoplus_{a=1}^N \mathbb{R}_a^3 \cong \mathbb{R}^{3N}$$

by permuting atoms and applying the spatial linear action to each displacement. The three-dimensional translational representation carries the polar-vector action R_g , while the rotational representation carries the axial-vector action $\det(R_g)R_g$; denote them by V_{trans} and V_{rot} . We complexify these real representations when applying character theory. For a nonlinear molecule, the vibrational representation is the class

$$[V_{\text{vib}}] = [V_{\text{disp}}] - [V_{\text{trans}}] - [V_{\text{rot}}]$$

in the representation ring.

Exercise 12 (Character of molecular displacements)

For $g \in G$, prove that

$$\chi_{\text{disp}}(g) = \sum_{\substack{a \\ g(a)=a}} \text{tr}(R_g),$$

where $R_g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the spatial action and the sum is over atoms fixed by g . Prove that $\dim V_{\text{vib}} = 3N - 6$ for a nonlinear molecule, and use character orthogonality to express the multiplicity of every irreducible representation in V_{vib} .

For water, take the z -axis along the C_2 axis and the molecule in the yz -plane. Its point group is

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Exercise 13 (Water has three symmetry-classified vibrations)

Use the group law and character orthogonality to derive the character table

	E	C_2	σ_{xz}	σ_{yz}	basis functions
A_1	1	1	1	1	z
A_2	1	1	-1	-1	R_z
B_1	1	-1	1	-1	x, R_y
B_2	1	-1	-1	1	y, R_x

Compute

$$\chi_{\text{disp}} = (9, -1, 1, 3), \quad \chi_{\text{trans}} = (3, -1, 1, 1), \quad \chi_{\text{rot}} = (3, -1, -1, -1).$$

Use character orthogonality to prove that

$$V_{\text{vib}} \cong 2A_1 \oplus B_2.$$

Identify the two A_1 modes as the symmetric stretch and bend, and the B_2 mode as the asymmetric stretch. Construct symmetry-adapted displacement vectors using the projections P_{A_1} and P_{B_2} .

1.5 Spectroscopic selection rules

Let V_i and V_f be the symmetry types of initial and final states. The electric-dipole operator transforms as the vector representation $V_\mu = V_{\text{trans}}$. A transition amplitude

$$\langle \psi_f, \mu \psi_i \rangle$$

is the matrix coefficient used below.

Exercise 14 (Symmetry determines electric-dipole transitions)

Use the Great Orthogonality Theorem to prove that the transition amplitude vanishes unless

$$1 \subseteq V_f^* \otimes V_\mu \otimes V_i.$$

For irreducible V_f , prove that this is equivalent to

$$V_f \subseteq V_\mu \otimes V_i.$$

The vibrational ground state of water has type A_1 . Since

$$V_\mu \cong A_1 \oplus B_1 \oplus B_2,$$

deduce that both A_1 modes and the B_2 mode are infrared active. Thus molecular symmetry alone predicts that all three fundamental vibrations of water are allowed in infrared spectroscopy.

Exercise 15 (All fundamental vibrations of water are Raman active)

Let the polarizability tensor carry $\text{Sym}^2(V_\mu)$. Prove that a vibrational mode is Raman active precisely when its irreducible representation occurs in this symmetric square. Prove that

$$\text{Sym}^2(V_\mu) \cong 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$$

for C_{2v} . Deduce that all three fundamental vibrational modes of water are Raman active.

1.6 Lie groups, infinitesimal symmetry, and spin

The molecular point groups above are discrete subgroups of $O(3)$. Define the group of orientation-preserving spatial rotations by

$$\mathrm{SO}(3) = \{R \in M_3(\mathbb{R}) : R^T R = 1, \det R = 1\}.$$

A **matrix Lie group** is a subgroup $G \subseteq \mathrm{GL}(n, \mathbb{C})$ that is closed in the subspace topology. Its **Lie algebra** is

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\},$$

where $e^{tX} = \sum_{k \geq 0} (tX)^k / k!$. The Lie bracket is $[X, Y] = XY - YX$. An abstract **Lie algebra** over k is a k -vector space with a bilinear bracket that is skew-symmetric and satisfies the Jacobi identity. A **Lie algebra representation** on V is a linear map $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ with $\rho([X, Y]) = [\rho(X), \rho(Y)]$.

For $g \in G$ and $X, Y \in \mathfrak{g}$, the adjoint actions are

$$\mathrm{Ad}_g(X) = gXg^{-1}, \quad \mathrm{ad}_X(Y) = [X, Y].$$

Exercise 16 (Classical matrix Lie algebras)

Prove that the Lie algebras of $\mathrm{GL}(n, \mathbb{C})$, $\mathrm{SL}(n, \mathbb{C})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(n)$ are respectively

$$\mathfrak{gl}(n, \mathbb{C}), \quad \mathfrak{sl}(n, \mathbb{C}) = \{X : \mathrm{tr}(X) = 0\},$$

$$\mathfrak{o}(n) = \{X : X^T = -X\}, \quad \mathfrak{u}(n) = \{X : X^* = -X\}, \quad \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n, \mathbb{C}).$$

Verify that each is a Lie algebra under $[X, Y] = XY - YX$.

Exercise 17 (One-parameter subgroups and the bracket)

Prove that $t \mapsto e^{tX}$ is a smooth group homomorphism $\mathbb{R} \rightarrow G$ whenever $X \in \mathfrak{g}$, and that every smooth homomorphism $\mathbb{R} \rightarrow G$ arises uniquely in this way. Prove that

$$[X, Y] = \left. \frac{d}{dt} \right|_{t=0} \left. \frac{d}{ds} \right|_{s=0} e^{tX} e^{sY} e^{-tX} e^{-sY},$$

and deduce $\mathrm{Ad}_{e^{tX}} = e^{t \mathrm{ad}_X}$.

Exercise 18 (Differentiating group representations)

Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a smooth finite-dimensional representation of a matrix Lie group. Prove that

$$\dot{\rho}(X) = \left. \frac{d}{dt} \right|_{t=0} \rho(e^{tX})$$

is a Lie algebra representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$, and that $\rho(e^{tX}) = e^{t \dot{\rho}(X)}$. If G is connected, prove that a subspace is G -invariant if and only if it is $\mathfrak{g}$ -invariant.

The **Pauli matrices** are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 19 (Rotations, angular momentum, and spin)

Identify $\mathfrak{so}(3)$ with $\mathbb{R}^3$ equipped with the cross-product bracket. Set $T_a = \sigma_a/2$ and $X_a = -iT_a$. Prove that X_1, X_2, X_3 form a basis of $\mathfrak{su}(2)$ and that

$$[X_a, X_b] = \sum_c \varepsilon_{abc} X_c, \quad [T_a, T_b] = i \sum_c \varepsilon_{abc} T_c.$$

Prove that the adjoint representation induces a double cover $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$. Classify the finite-dimensional irreducible complex representations V_j of $\mathrm{SU}(2)$. For the unitary representation ρ_j , set $J_a = i\rho_j(X_a)$; prove that $\dim V_j = 2j + 1$ and that $\sum_a J_a^2 = j(j + 1)1_{V_j}$, and explain why integer spin descends to $\mathrm{SO}(3)$ while half-integer spin gives projective representations of spatial rotations.

A **quantum number** is an eigenvalue used to label a joint eigenspace of commuting observables.

Exercise 20 (Energy and symmetry produce quantum numbers)

Let H be a self-adjoint operator on a finite-dimensional Hermitian space V , and let $\rho : \mathrm{SU}(2) \rightarrow U(V)$ commute with H . Prove that every spectral subspace of H is $\mathrm{SU}(2)$ -invariant and decomposes into irreducible representations. For $J_a = i\rho(X_a)$ and $J^2 = \sum_a J_a^2$, prove that V has an orthonormal basis

$$|E, j, m, \alpha\rangle$$

satisfying

$$H|E, j, m, \alpha\rangle = E|E, j, m, \alpha\rangle,$$

$$J^2|E, j, m, \alpha\rangle = j(j + 1)|E, j, m, \alpha\rangle, \quad J_3|E, j, m, \alpha\rangle = m|E, j, m, \alpha\rangle.$$

Determine the allowed values of j and m , identify the multiplicity label α , and prove that the occurrence of V_j in an energy eigenspace forces degeneracy at least $2j + 1$. Interpret E, j, m, α as quantum numbers.

1.7 Monoidal and braided categories

A $\mathbb{C}$ -linear category is a category whose morphism sets

$$\mathrm{Hom}_{\mathcal{C}}(x, y)$$

are complex vector spaces and whose composition is bilinear.

A $\mathbb{C}$ -linear monoidal category is a $\mathbb{C}$ -linear category equipped with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1 , up to coherent natural isomorphism.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\mathrm{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple

objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities.

The definitions of duality, trace, and dimension above extend verbatim to $\mathcal{C}$. Write $d_x = \text{tr}(1_x)$. Assume that the categorical dimensions of nonzero objects are positive real numbers.

Exercise 21 (Categorical trace and dimension)

Repeat the first three parts of Exercise 2 in $\mathcal{C}$.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \rightarrow x'$ and $g : y \rightarrow y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$\begin{aligned} c_{x,y \otimes z} &= (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z), \\ c_{x \otimes y,z} &= (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}). \end{aligned}$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\text{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$. Assume first that every simple object is invertible and that their isomorphism classes form the cyclic group

$$\{1, a, a^2, \dots, a^{m-1}\}, \quad a^r \otimes a^s \cong a^{r+s \pmod{m}}.$$

Write $U(1) = \{z \in \mathbb{C} : |z| = 1\}$ and $M_{r,s} = M_{a^r, a^s} \in U(1)$.

Exercise 22 (The $\nu = 1/3$ anyonic phase advances by $2\pi/3$)

Use the hexagon identities to prove that

$$M_{r+s,t} = M_{r,t}M_{s,t}, \quad M_{r,s+t} = M_{r,s}M_{r,t}.$$

Deduce that there is an integer k modulo m such that

$$M_{r,s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $k = 1$, prove that the monodromy of a about N copies of a is multiplication by

$$\exp\left(\frac{2\pi i N}{m}\right).$$

In particular, for $m = 3$, increasing N by one changes the phase by $2\pi/3$. Compare this conclusion with the interferometric observation of the $\nu = 1/3$ fractional quantum Hall state in [23].

Now suppose that $\mathcal{C}$ has two simple objects $1, \tau$ satisfying the fusion rule

$$\tau \otimes \tau \cong 1 \oplus \tau.$$

Exercise 23 (Fibonacci categorical dimension)

Let $d_\tau = \text{tr}(1_\tau)$. Prove that

$$d_\tau^2 = 1 + d_\tau$$

and, using positivity, that

$$d_\tau = \frac{1 + \sqrt{5}}{2}.$$

Let

$$V_n = \text{Hom}_{\mathcal{C}}(1, \tau^{\otimes n}).$$

Show that the dimensions of these fusion spaces satisfy the Fibonacci recurrence and determine their asymptotic growth rate.

Let

$$V = \text{Hom}_{\mathcal{C}}(\tau, \tau^{\otimes 3}).$$

Set $\varphi = (1 + \sqrt{5})/2$. In the basis indexed by the intermediate labels $1, \tau$, define [22]

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

Define $\sigma_1 = R$ and $\sigma_2 = F^{-1}RF$. Let

$$B_3 = \langle b_1, b_2 \mid b_1 b_2 b_1 = b_2 b_1 b_2 \rangle.$$

Exercise 24 (Fibonacci anyons braid non-Abelianly)

Use the fusion rule to prove that V is two-dimensional with basis indexed by the intermediate objects $1, \tau$ in $(\tau \otimes \tau) \otimes \tau$. Working exactly, verify that F and R are unitary and prove that

$$\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2.$$

Prove also that $\sigma_1 \sigma_2 \neq \sigma_2 \sigma_1$. Deduce that $b_i \mapsto \sigma_i$ defines a non-Abelian representation

$$B_3 \longrightarrow U(V).$$

Relate these fusion and braiding data to the engineered superconducting-processor realization in [25] and the related trapped-ion realization in [24]; distinguish these simulations of non-Abelian topological order from evidence for naturally occurring Fibonacci quasiparticles.

2 Geometry

2.1 Differential forms and cohomology

A **smooth manifold** is a topological space equipped with an atlas whose transition maps are smooth. A smooth map $f : M \rightarrow N$ has a tangent map $df : TM \rightarrow TN$. The **tangent bundle** TM is the union of the tangent spaces, the **cotangent bundle** is T^*M , and a **vector field** is a smooth section of TM . An orientation of M is a smoothly varying choice of orientation of its tangent spaces, using the determinant lines defined in Algebra.

The exterior powers $\Lambda^k T_p^*M$ are those defined in the Algebra section. To define integration in a coordinate-free way, we will integrate differential forms over chains. The space of **differential n -forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^*M)$. On an oriented n -manifold, a **volume form** is a nowhere-vanishing element of $\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n -simplex** is a smooth map $\sigma : \Delta^n \rightarrow M$ where Δ^n is the standard n -simplex. The collection of singular n -simplices in M freely generates the abelian group $C_n(M)$ of **singular n -chains** in M . The **boundary map** $\partial : C_n(M) \rightarrow C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \rightarrow M$ by

$$\partial\sigma = \sum_{j=0}^n (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \rightarrow \Delta^n$ is the inclusion of the face opposite the j -th vertex of Δ^n .

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \rightarrow A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n . The n -th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n) / \operatorname{im}(d_{n+1}).$$

The prototypical example is singular homology which corresponds to the chain complex of singular chains with the boundary map defined above.

Cohomology and cochains are defined similarly, using superscripts instead of subscripts and boundary maps $d^n : A^n \rightarrow A^{n+1}$ increase degree instead of decrease it. Singular cohomology corresponds to the cochain complex given by dualizing the singular chain complex above.

The **de Rham cohomology** of M is

$$H_{\text{dR}}^n(M) = \ker(d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)) / \operatorname{im}(d : \Omega^{n-1}(M) \rightarrow \Omega^n(M)).$$

For a smooth map $f : N \rightarrow M$, the **pullback** of $\alpha \in \Omega^p(M)$ is

$$(f^*\alpha)_x(v_1, \dots, v_p) = \alpha_{f(x)}(df_x v_1, \dots, df_x v_p).$$

Exercise 25 (Poincaré lemma and Stokes' theorem)

Prove that $(f \circ g)^* = g^* f^*$, that pullback preserves wedge products, and that $d(f^* \alpha) = f^*(d\alpha)$. Prove that $d^2 = 0$. Prove the Poincaré lemma: every closed form is locally exact in positive degree. Define the integral of an n -form on an n -chain and prove

$$\int_c d\omega = \int_{\partial c} \omega.$$

Exercise 26 (de Rham theorem)

Prove that integration over smooth singular cycles induces a natural isomorphism between de Rham cohomology and singular cohomology with real coefficients.

Exercise 27 (Nontrivial cohomology produces the Aharonov–Bohm phase)

Let M be a smooth manifold and let $A \in \Omega^1(M)$ satisfy $dA = 0$. Prove the following:

- For every contractible open set $U \subseteq M$, there exists $\lambda \in C^\infty(U)$ such that $A|_U = d\lambda$.
- For every smooth closed curve γ , the period $\int_\gamma A$ depends only on the homology class of γ .
- There exists a global $\lambda \in C^\infty(M)$ with $A = d\lambda$ if and only if $[A] = 0 \in H_{\text{dR}}^1(M)$. If $[A] \neq 0$, use the de Rham theorem to find a smooth closed curve with nonzero period.

Now take $M = \mathbb{R}^3 \setminus \{(0, 0, z) : z \in \mathbb{R}\}$, the space outside an idealized solenoid, and define

$$d\theta = \frac{-y dx + x dy}{x^2 + y^2}, \quad A = \frac{\Phi}{2\pi} d\theta.$$

Prove that M deformation retracts onto S^1 , that $d\theta$ is closed but not exact, and that its integral around a positively oriented loop linking the solenoid once is 2π . Deduce that $\int_\gamma A = \Phi$. For a particle of charge q , prove that

$$\exp\left(\frac{iq}{\hbar} \int_\gamma A\right)$$

is unchanged by $A \mapsto A + d\lambda$, and explain why the relative phase $q\Phi/\hbar$ shifts the interference pattern although $dA = 0$ on M [21].

Exercise 28 (Categorical trace on chain complexes)

Prove that bounded chain complexes of finite-dimensional vector spaces form a symmetric monoidal abelian category with duals, that the categorical trace of an endomorphism is its Lefschetz number, and that the categorical dimension of a chain complex is its Euler characteristic.

2.2 Hodge theory and electromagnetism

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n -manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. A **metric volume form** vol_g is a positive volume form satisfying

$$|\langle \text{vol}_g, \text{vol}_g \rangle_g| = 1.$$

The **Hodge star** is an operator $\star : \Omega^p(M) \rightarrow \Omega^{n-p}(M)$ satisfying

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_g \text{vol}_g.$$

Exercise 29 (Metric volume and Hodge star)

Prove that an oriented pseudo-Riemannian manifold has a unique metric volume form and a unique Hodge-star operator. Determine $\star^2$ in terms of the degree and signature of the metric.

On an oriented Lorentzian four-manifold, the electromagnetic field strength is a two-form F and the electric four-current is a three-form J . Maxwell's equations are

$$dF = 0, \quad d \star F = J.$$

Exercise 30 (Maxwell's equations imply charge conservation)

Prove that Maxwell's equations imply $dJ = 0$. If C is an oriented compact four-dimensional chain, deduce that

$$\int_{\partial C} J = 0.$$

If the three-dimensional pieces of its boundary satisfy

$$\partial C = \Sigma_1 - \Sigma_0 + B,$$

prove that

$$\int_{\Sigma_1} J - \int_{\Sigma_0} J = - \int_B J.$$

Specialize to Minkowski spacetime $\mathbb{R}_t \times \mathbb{R}^3$ with metric $-dt^2 + dx^2 + dy^2 + dz^2$. Let $\star_3$ and vol_3 be the Euclidean Hodge star and volume form on space, and write

$$F = B - dt \wedge E, \quad J = \rho \text{vol}_3 - dt \wedge \star_3 j,$$

where E, j are spatial one-forms and B is a spatial two-form. Prove that Maxwell's equations are equivalent to

$$d_3 B = 0, \quad \partial_t B + d_3 E = 0,$$

$$d_3 \star_3 E = \rho \text{vol}_3, \quad d_3 \star_3 B - \partial_t \star_3 E = \star_3 j.$$

Identify these with Gauss's laws, Faraday's law, and the Ampère–Maxwell law. Show that $dJ = 0$ becomes the continuity equation

$$\partial_t \rho \text{vol}_3 + d_3 \star_3 j = 0.$$

Exercise 31 (Gauge invariance forces charge conservation)

Assume that M has no boundary, or restrict to compactly supported variations. For $A \in \Omega^1(M)$, set $F = dA$ and define

$$S[A] = -\frac{1}{2} \int_M F \wedge \star F + \int_M A \wedge J.$$

For $\alpha \in \Omega^1(M)$, compute the first variation of $S[A + t\alpha]$ and prove that its Euler–Lagrange equation is $d \star F = J$. Explain why $dF = 0$ holds identically. Prove that the action is invariant under $A \mapsto A + d\lambda$ when $dJ = 0$, and show conversely how gauge invariance of the source term forces charge conservation.

2.3 Bundles, K-theory, and band topology

A **smooth vector bundle** $p : E \rightarrow M$ is a smooth family of vector spaces locally isomorphic to a product $U \times V \rightarrow U$. For a smooth map $f : N \rightarrow M$, its **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \rightarrow N.$$

For a compact Hausdorff space X , let $\text{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X , with the usual direct sum, tensor product, dual, and pullback constructions. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E \oplus F] = [E] + [F].$$

Define multiplication by $[E][F] = [E \otimes F]$. For a continuous map $f : X \rightarrow Y$ between compact Hausdorff spaces, define pullback by

$$f^* : K^0(Y) \rightarrow K^0(X), \quad [E] \mapsto [f^*E].$$

Exercise 32 (Grothendieck group of vector bundles)

Prove that $(f \circ g)^*E \cong g^*f^*E$ and that pullback preserves direct sums, tensor products, and duals. Let X and Y be compact Hausdorff spaces and let $f : X \rightarrow Y$ be continuous. Prove that direct sum and tensor product induce a commutative ring structure on $K^0(X)$ and that f^* is a ring homomorphism. Prove that

$$K^0(\text{pt}) \cong \mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0 \in X$, regarded as a map $x_0 : \text{pt} \rightarrow X$. The **reduced K-theory** of the pointed space (X, x_0) is

$$\tilde{K}^0(X) = \ker \left(K^0(X) \xrightarrow{x_0^*} K^0(\text{pt}) \right).$$

For a bundle E of rank r , its **reduced class** is

$$[E] - [\mathbb{C}^r],$$

where $\underline{\mathbb{C}}^r = X \times \mathbb{C}^r$ is the trivial rank- r bundle. Write $K^n(X)$ for the generalized cohomology groups extending $K^0(X)$.

Exercise 33 (Stable equivalence and Bott periodicity)

Let E and F be complex vector bundles over a compact Hausdorff space X . Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \underline{\mathbb{C}}^n$. Prove Bott periodicity

$$K^{n+2}(X) \cong K^n(X)$$

and deduce that complex K -theory is naturally $\mathbb{Z}/2$ -graded.

For $0 \leq r \leq N$, identify the complex Grassmannian $\text{Gr}_r(\mathbb{C}^N)$ with the space of rank- r orthogonal projections in $M_N(\mathbb{C})$. Its **tautological bundle** has fiber $\text{im } P$ over P . Let BU denote the stable complex Grassmannian obtained from finite Grassmannians by adjoining trivial summands.

Exercise 34 (Stable classification of complex vector bundles)

Let X be compact Hausdorff. Use a finite trivializing cover and a partition of unity to prove that every complex vector bundle $E \rightarrow X$ is a direct summand of a trivial bundle: there are a bundle $E^\perp$ and an integer N such that

$$E \oplus E^\perp \cong \underline{\mathbb{C}}^N.$$

After choosing a Hermitian metric, deduce that E is the image bundle of a continuous projection

$$P_E : X \longrightarrow \text{Gr}_r(\mathbb{C}^N) \subseteq M_N(\mathbb{C}).$$

Conversely, prove that every continuous projection-valued map has a vector bundle as its image. Prove that a homotopy of projections determines a vector bundle over $X \times [0, 1]$ and hence gives isomorphic image bundles at its endpoints. Prove the stable converse: two bundles have the same reduced K -theory class exactly when, after adjoining trivial bundles, their projection-valued maps become homotopic.

For a connected finite CW complex X , deduce the natural classifications

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \quad \tilde{K}^0(X) \cong [X, BU]_*.$$

Here the first integer records virtual rank and the star denotes based homotopy classes. Explain how direct sum of bundles corresponds to the H -space addition on $\mathbb{Z} \times BU$.

For the gapped families below, assume that X is connected, so vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space as

in Algebra. Exercise 6 supplies the spectral decomposition used below. A continuous family is a map

$$H : X \longrightarrow \text{End}(V)$$

and write $H_x = H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Given a positively oriented contour Γ enclosing the negative spectra and excluding the positive spectra, define

$$P_x^- = \frac{1}{2\pi i} \int_{\Gamma} (z - H_x)^{-1} dz, \quad E_x^- = \text{im}(P_x^-).$$

Exercise 35 (Negative spectral bundle)

Let H be a gapped family over compact X . Prove that there are $\delta, M > 0$ such that every eigenvalue of every H_x lies in $[-M, -\delta] \cup [\delta, M]$, and hence that a contour Γ as above exists. Prove that

$$(P_x^-)^2 = P_x^-, \quad (P_x^-)^* = P_x^-,$$

that $x \mapsto P_x^-$ is continuous, and that

$$E_H^- = \coprod_{x \in X} E_x^-$$

is a complex vector bundle over X . Prove that its rank is locally constant.

For a gapped family $H : X \times [0, 1] \rightarrow \text{End}(V)$, let E_j^- be the negative spectral bundle of $x \mapsto H(x, j)$, and let $i_j : X \rightarrow X \times [0, 1]$ be given by $i_j(x) = (x, j)$.

Exercise 36 (Gap-preserving homotopy)

Prove that the negative spectral subspaces of H form a vector bundle $E^- \rightarrow X \times [0, 1]$. Prove that the endpoint pullbacks of any vector bundle over $X \times [0, 1]$ are isomorphic, and apply this to $i_0^* E^-$ and $i_1^* E^-$. Deduce that

$$[E_0^-] = [E_1^-] \in K^0(X).$$

Let $Q \in \text{End}(W)$ be a fixed invertible self-adjoint endomorphism of a finite-dimensional Hermitian space W , let W^- be its negative eigenspace, and write $\underline{W}^- = X \times W^-$.

Exercise 37 (Stabilization)

For a gapped family H , prove that

$$E_{H \oplus Q}^- \cong E_H^- \oplus \underline{W}^-.$$

Deduce that

$$[E_H^-] - [\mathbb{C}^{\text{rank } E_H^-}] \in \tilde{K}^0(X)$$

is unchanged by adjoining a constant gapped summand.

Let $E \subseteq X \times V$ be a subbundle and let P_x be the orthogonal projection onto E_x .

Exercise 38 (Flattened family)

Define $H_x = 1_V - 2P_x$. Prove that $x \mapsto H_x$ is a gapped family, that

$$H_x^2 = 1_V,$$

and that its negative spectral bundle is E . Deduce that every subbundle of a trivial bundle arises as the negative spectral bundle of a gapped family. Combine the preceding exercises to prove that a finite-dimensional gapped family determines a class in $\tilde{K}^0(X)$ invariant under gap-preserving homotopy and stabilization.

Consider a translation-invariant noninteracting band insulator in symmetry class A: there is no time-reversal, particle-hole, or chiral symmetry, and its Bloch Hamiltonian is a smooth gapped family

$$H : \mathbb{T}^d \longrightarrow \text{End}(V)$$

over the Brillouin torus. The negative-energy bundle E_H^- is the bundle of occupied states. Declare two such Hamiltonians **stably equivalent** when they become connected by a smooth gap-preserving homotopy after adjoining momentum-independent filled or empty bands [1].

Exercise 39 (K-theory classifies class A band insulators)

Use the preceding exercises and Exercise 34 to prove that

$$H \longmapsto [E_H^-] - [\mathbb{C}^{\text{rank } E_H^-}]$$

induces a bijection from stable equivalence classes of class A Hamiltonians over a compact connected space X to $\tilde{K}^0(X)$. In particular, show that every reduced K-theory class is realized by a flattened Hamiltonian $H = 1 - 2P$.

Using the suspension isomorphism and Bott periodicity, prove that the strong class A phases of continuum models whose momentum space compactifies to S^d form

$$\tilde{K}^0(S^d) \cong \begin{cases} \mathbb{Z}, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Using the Künneth theorem and the graded ring $K^*(S^1)$, prove that

$$K^0(\mathbb{T}^d) \cong \bigoplus_{\substack{0 \leq j \leq d \\ j \text{ even}}} \mathbb{Z}^{\binom{d}{j}}, \quad \tilde{K}^0(\mathbb{T}^d) \cong \bigoplus_{\substack{2 \leq j \leq d \\ j \text{ even}}} \mathbb{Z}^{\binom{d}{j}}.$$

Deduce that the reduced classifications over $\mathbb{T}^d$ for $d = 1, 2, 3$ are respectively 0, $\mathbb{Z}$, and $\mathbb{Z}^3$. Distinguish the strong invariant from the weak invariants inherited from lower-dimensional subtori. Explain why this is not the full tenfold-way classification: antiunitary symmetries impose Real or Quaternionic structures on the occupied states and require the corresponding variants of K-theory [1].

2.4 Symplectic geometry

For a vector field X , contraction is the map $\iota_X : \Omega^p(M) \rightarrow \Omega^{p-1}(M)$ defined by

$$(\iota_X \alpha)(X_2, \dots, X_p) = \alpha(X, X_2, \dots, X_p).$$

Define the Lie derivative of forms by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X \omega$ is a bundle isomorphism $TM \rightarrow T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M . For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H} \omega = dH.$$

Let $\pi : T^*Q \rightarrow Q$ be the cotangent bundle of a configuration manifold. Its **tautological one-form** is

$$\theta_{(q,p)}(\xi) = p(d\pi_{(q,p)}\xi),$$

and its **canonical symplectic form** is $\omega_0 = -d\theta$.

Exercise 40 (The symplectic form produces Hamilton's equations)

Prove that the definition of θ is coordinate independent. In local cotangent coordinates (q^i, p_i) , prove that

$$\theta = \sum_{i=1}^n p_i dq^i, \quad \omega_0 = \sum_{i=1}^n dq^i \wedge dp_i,$$

and prove that ω_0 is symplectic. For

$$H(q, p) = \frac{1}{2m} \sum_{i=1}^n p_i^2 + V(q),$$

derive Hamilton's equations

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i},$$

and hence Newton's equation $m\ddot{q} = -\nabla V(q)$. For $V(q) = m\Omega^2|q|^2/2$, solve the flow and prove that every nonconstant trajectory has period $2\pi/\Omega$. In the one-dimensional case, describe the constant-energy curves as ellipses in phase space.

Exercise 41 (Magnetic symplectic curvature produces the Lorentz force)

Let $Q = \mathbb{R}^3$, let $B \in \Omega^2(Q)$ satisfy $dB = 0$, and let e be the charge of a particle. Prove that the magnetically twisted form

$$\omega_B = \omega_0 - e\pi^*B$$

is symplectic on T^*Q . For a static electric potential ϕ and Hamiltonian

$$H(q, p) = \frac{|p|^2}{2m} + e\phi(q),$$

identify B with the vector field $\mathbf{B}$ determined by $B(u, v) = \mathbf{B} \cdot (u \times v)$ and derive the Lorentz-force equation

$$m\ddot{q} = e(\mathbf{E} + \dot{q} \times \mathbf{B}), \quad \mathbf{E} = -\nabla\phi.$$

If $B = dA$, prove that $\omega_B = -d(\theta + e\pi^*A)$ and that $A \mapsto A + d\lambda$ leaves ω_B and the equations of motion unchanged. Relate this gauge freedom to the potential used in the Aharonov–Bohm exercise.

Exercise 42 (Liouville’s theorem)

Let (M, ω) be $2n$ -dimensional with ω symplectic. Prove that $\omega^n/n!$ is a volume form. For $H \in C^\infty(M)$, prove using Cartan’s formula that

$$\mathcal{L}_{X_H}\omega = 0.$$

Deduce that $\mathcal{L}_{X_H}\omega^n = 0$ and hence that the local flow of X_H preserves the volume form

$$\frac{\omega^n}{n!}.$$

Fix a phase-volume unit so that the logarithm of a phase-space density is dimensionless. For a probability density ρ with respect to Liouville measure $d\mu_L = \omega^n/n!$, define the **fine-grained Gibbs entropy** by

$$S_G[\rho] = -k_B \int_M \rho \log \rho d\mu_L.$$

Exercise 43 (Hamiltonian phase volume produces ideal-gas thermodynamics)

Let Φ_t be a Hamiltonian flow and let $\rho_t = \rho_0 \circ \Phi_{-t}$. Use Liouville’s theorem to prove that ρ_t remains normalized and that

$$S_G[\rho_t] = S_G[\rho_0].$$

Now consider N indistinguishable noninteracting particles of mass m in a three-dimensional box of volume V , with Hamiltonian

$$H(q, p) = \sum_{a=1}^N \frac{|p_a|^2}{2m}.$$

Divide phase volume by $N!h^{3N}$ and prove that the volume of the region $H \leq E$ is

$$\Omega(E, V, N) = \frac{V^N}{N!h^{3N}} \frac{(2\pi m E)^{3N/2}}{\Gamma(3N/2 + 1)}.$$

Define the microcanonical entropy and its thermodynamic derivatives by

$$S_{\text{mc}} = k_B \log \Omega, \quad \frac{1}{T} = \left(\frac{\partial S_{\text{mc}}}{\partial E} \right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S_{\text{mc}}}{\partial V} \right)_{E,N}.$$

Derive

$$E = \frac{3}{2} N k_B T, \quad PV = N k_B T.$$

Use Stirling's formula to obtain the Sackur–Tetrode expression

$$S_{\text{mc}} = N k_B \left[\log \left(\frac{V}{N} \left(\frac{4\pi m E}{3N h^2} \right)^{3/2} \right) + \frac{5}{2} \right] + o(N).$$

For a reversible adiabatic process, set $dS_{\text{mc}} = 0$ and prove

$$PV^{5/3} = \text{constant}.$$

If $\varrho = mN/V$ is the mass density, use $c^2 = (\partial P / \partial \varrho)_S$ to derive the measurable sound-speed law

$$c = \sqrt{\frac{5P}{3\varrho}}.$$

Finally, show that removing a partition and allowing a free expansion from V to $2V$ changes the thermodynamic entropy by $N k_B \log 2$. Explain why this does not contradict conservation of the fine-grained Gibbs entropy: the former compares equilibrium macrostates after changing the accessible region, whereas the latter follows the exact Hamiltonian density without coarse-graining.

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

Exercise 44 (Hamiltonian symmetry produces conservation laws)

Let (M, ω) be symplectic. Prove that

$$X_g(f) = \{f, g\}.$$

Let Φ_t be the local flow of X_f . Prove that the following are equivalent:

- $H \circ \Phi_t = H$ whenever both sides are defined.
- $X_f(H) = 0$.
- $\{H, f\} = 0$.
- f is constant along the integral curves of X_H .

Explain why this is a Hamiltonian symmetry–conservation statement rather than the full variational theorem of [13].

Exercise 45 (Central forces and angular momentum)

On $T^*\mathbb{R}^3$, let

$$H(q, p) = \frac{|p|^2}{2m} + V(|q|), \quad L = q \times p.$$

Prove that the components of angular momentum satisfy

$$\{L_i, L_j\} = \sum_k \varepsilon_{ijk} L_k, \quad \{L_i, H\} = 0.$$

Deduce from the Hamiltonian Noether theorem that L is conserved. Show that when $L \neq 0$ the trajectory remains in the fixed plane through the origin perpendicular to L , and interpret this conservation law as a consequence of rotational symmetry of the central potential.

2.5 Connections and curvature

An **affine connection** is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM)$ such that

$$\nabla_{fX} Y = f \nabla_X Y, \quad \nabla_X (fY) = X(f)Y + f \nabla_X Y.$$

For a covector field α , set

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

Define the covariant derivative of general tensor fields by the product rule and by requiring it to commute with contractions.

A connection is **torsion-free and metric-compatible** when

$$\nabla_X Y - \nabla_Y X = [X, Y], \quad \nabla g = 0.$$

Such a connection is called a **Levi-Civita connection**.

Exercise 46 (Levi-Civita connection)

Prove that every pseudo-Riemannian manifold admits a unique Levi-Civita connection. Derive the coordinate-free Koszul formula.

An affinely parametrized curve $\gamma : I \rightarrow M$ is a **geodesic** if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_X g)(Y, Z) = X(g(Y, Z)) - g([X, Y], Z) - g(Y, [X, Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_X g = 0$.

Let (M, g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇ .

Exercise 47 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt} g(\dot{\gamma}, \dot{\gamma}) = 0, \quad \frac{d}{dt} g(X, \dot{\gamma}) = 0.$$

Deduce that $g(X, \dot{\gamma})$ is constant along γ .

The **curvature** of ∇ is the $\text{End}(TM)$ -valued 2-form F_∇ determined by

$$R(X, Y)Z := F_\nabla(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z.$$

Exercise 48 (Curvature as a tensor)

For $f \in C^\infty(M)$ and vector fields X, Y, Z , prove that

$$R(fX, Y)Z = fR(X, Y)Z, \quad R(X, fY)Z = fR(X, Y)Z,$$

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

Prove that $R(X, Y) = -R(Y, X)$ and deduce that

$$R \in \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes TM).$$

Let $\Gamma : (-\varepsilon, \varepsilon) \times I \rightarrow M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by

$$T = \frac{\partial \Gamma}{\partial t}, \quad J = \frac{\partial \Gamma}{\partial s}.$$

Exercise 49 (Geodesic deviation) For a variation Γ as above, prove that $[T, J] = 0$ and that

$$\nabla_T \nabla_T J = R(T, J)T.$$

Let (M, g) be a compact oriented Riemannian surface with piecewise smooth boundary. Given a local positively oriented orthonormal frame (e_1, e_2) , define its **connection one-form** ω by

$$\nabla_X e_1 = -\omega(X)e_2, \quad \nabla_X e_2 = \omega(X)e_1.$$

The **Gaussian curvature** is

$$K = g(R(e_1, e_2)e_2, e_1),$$

and the Euler characteristic is $\chi(M) = \sum_j (-1)^j \dim H_j(M; \mathbb{R})$. Along a positively oriented smooth boundary component with unit tangent T , let N be the inward unit normal for which (T, N) is positively oriented and define the signed **geodesic curvature** by $k_g = g(\nabla_T T, N)$.

Exercise 50 (Gauss–Bonnet theorem)

Prove that [14]

$$d\omega = K \text{vol}_g.$$

Determine how ω transforms under a rotation of the local orthonormal frame and deduce that $K \text{vol}_g$ is globally defined. Using Stokes' theorem on a triangulation, let θ_j be the signed exterior angle through which the positively oriented boundary tangent turns at its j -th corner, and prove the Gauss–Bonnet formula

$$\int_M K \text{vol}_g + \int_{\partial M} k_g ds + \sum_j \theta_j = 2\pi \chi(M).$$

Deduce that every closed oriented Riemannian surface satisfies

$$\frac{1}{2\pi} \int_M K \operatorname{vol}_g = \chi(M).$$

Apply the theorem to recover the area $4\pi r^2$ of the round sphere of radius r . Prove that a torus admits no metric with $K > 0$ everywhere, and that a closed genus- g surface with constant curvature $K = -1$ has area $4\pi(g-1)$.

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\operatorname{Ric}(X, Y) = \operatorname{tr}(Z \mapsto R(Z, X)Y), \quad S = \operatorname{tr}_g(\operatorname{Ric}), \quad G = \operatorname{Ric} - \frac{1}{2}Sg.$$

For a symmetric covariant two-tensor A , its **divergence** is the covector field

$$(\operatorname{div} A)(Y) = \operatorname{tr}_g((X, Z) \mapsto (\nabla_X A)(Z, Y)).$$

Exercise 51 (Contracted Bianchi identity and conservation)

Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\operatorname{div} G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\operatorname{div} T = 0$.

Let $E \rightarrow M$ be a real or complex vector bundle and write $\Omega^0(M, E) = \Gamma(E)$. A **connection** on E is a map

$$\nabla : \Omega^0(M, E) \longrightarrow \Omega^1(M, E)$$

linear over the ground field and satisfying $\nabla(fs) = df \otimes s + f\nabla s$. For $\alpha \in \Omega^p(M)$ and an E -valued form η , extend it by

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^p \alpha \wedge \nabla \eta.$$

Define the curvature by $F_\nabla = \nabla^2 \in \Omega^2(M, \operatorname{End} E)$, and equip $\operatorname{End} E$ -valued forms with the induced connection.

Exercise 52 (Bundle curvature agrees with Riemann curvature and satisfies Bianchi)

For $E = TM$, prove that $F_\nabla = \nabla^2$ agrees with the Riemann-curvature formula above. Let ∇ be a connection on $E \rightarrow M$. Prove that

$$\nabla F_\nabla = 0.$$

Let $H : X \rightarrow \text{End}(V)$ be a smooth family of self-adjoint operators with a simple isolated eigenvalue λ_x , and set

$$L = \{(x, v) \in X \times V : H_x v = \lambda_x v\}.$$

For a local unit eigenvector ψ , define the **Berry connection** and **Berry curvature** by

$$\mathcal{A} = i\langle\psi, d\psi\rangle, \quad \mathcal{F} = d\mathcal{A}.$$

Exercise 53 (Berry phase as holonomy)

Prove that $L \rightarrow X$ is a complex line bundle. Under the change $\psi \mapsto e^{i\lambda}\psi$, prove that $\mathcal{A} \mapsto \mathcal{A} - d\lambda$ and that $\mathcal{F}$ is unchanged. For a closed curve γ in X , prove that

$$\exp\left(i \int_{\gamma} \mathcal{A}\right)$$

is the holonomy of the eigenline connection and is independent of the local choice of eigenvector. Interpret it as the geometric phase accumulated during adiabatic transport.

Let P be an invariant symmetric k -linear polynomial on the endomorphism algebra of a fiber of E , where invariant means invariant under simultaneous conjugation. Define

$$P(F_{\nabla}) = P(F_{\nabla}, \dots, F_{\nabla}) \in \Omega^{2k}(M).$$

Exercise 54 (Chern–Weil theorem)

Prove that

$$dP(F_{\nabla}) = 0.$$

For two connections ∇^0, ∇^1 on E , prove that

$$[P(F_{\nabla^0})] = [P(F_{\nabla^1})] \in H_{\text{dR}}^{2k}(M).$$

Relate this construction to the characteristic forms of [15].

Define the **Chern character form** by

$$\text{ch}(E, \nabla) = \text{tr} \exp\left(\frac{iF_{\nabla}}{2\pi}\right).$$

When induced connections are understood, suppress them from the notation.

Exercise 55 (Chern character)

Prove that $d \text{ch}(E, \nabla) = 0$ and that its de Rham class is independent of ∇ . For a second complex vector bundle F with connection, prove that

$$\text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F),$$

$$\text{ch}(E \otimes F) = \text{ch}(E) \wedge \text{ch}(F).$$

If M is compact, deduce that the Chern character defines a ring homomorphism

$$\text{ch} : K^0(M) \longrightarrow H_{\text{dR}}^{\text{even}}(M), \quad H_{\text{dR}}^{\text{even}}(M) = \bigoplus_{j \geq 0} H_{\text{dR}}^{2j}(M).$$

Return to the two-dimensional class A insulator in Exercise 39. For its smooth occupied-state projector P , prove that the Grassmann connection $\nabla = P d$ has curvature $F_{\nabla} = P dP \wedge dP$ on E_H^- . Show that the integer classifying the phase is the first Chern number

$$C_1(E_H^-) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \text{tr}(P dP \wedge dP) \in \mathbb{Z},$$

and deduce that a nonzero Chern number obstructs a global smooth frame of occupied Bloch states.

2.6 Yang–Mills theory and the Standard Model

Let G be a Lie group with Lie algebra $\mathfrak{g}$. A **principal G -bundle** is a smooth free right G -manifold P with quotient $M = P/G$ such that the projection $P \rightarrow M$ is a locally trivial fiber bundle. For a finite-dimensional representation $\rho : G \rightarrow \text{GL}(V)$, the **associated bundle** is

$$E = P \times_G V = (P \times V) / \sim, \quad (p \cdot g, v) \sim (p, \rho(g)^{-1}v).$$

A **principal connection** is a $\mathfrak{g}$ -valued one-form A on P that is G -equivariant and reproduces the generators of the fundamental vector fields. In a local trivialization it is represented by an ordinary form $A \in \Omega^1(U, \mathfrak{g})$, with curvature

$$F_A = dA + A \wedge A \in \Omega^2(U, \mathfrak{g}).$$

Exercise 56 (Isospin motivates Yang–Mills connections)

Treat the proton and neutron as a nucleon doublet carrying the defining representation of $\text{SU}(2)$. Determine the obstruction to replacing its global $\text{SU}(2)$ symmetry by a spacetime-dependent transformation $g : U \rightarrow \text{SU}(2)$. Prove that a covariant derivative requires a local connection form transforming by

$$A \longmapsto A^g = g^{-1} A g + g^{-1} dg.$$

Show that a principal connection A induces a covariant derivative

$$\nabla_A : \Gamma(E) \longrightarrow \Gamma(T^*M \otimes E)$$

on every associated vector bundle $E = P \times_G V$. Prove that under a gauge transformation the curvature transforms by

$$F_{A^g} = g^{-1} F_A g.$$

Relate this construction to the original non-Abelian gauge theory of isospin in [16].

On an oriented Riemannian four-manifold, extend the Hodge star to $\mathfrak{g}$ -valued forms using a fixed Ad-invariant inner product on $\mathfrak{g}$, written Tr . The Yang–Mills action of a connection is

$$S(A) = \int_M \text{Tr}(F_A \wedge \star F_A).$$

Write $d_A \alpha = d\alpha + [A, \alpha]$ for the exterior covariant derivative on $\mathfrak{g}$ -valued forms.

Exercise 57 (Yang–Mills action)

Prove that $S(A)$ is invariant under gauge transformations. Computing the first variation, derive the Euler–Lagrange equation

$$d_A \star F_A = 0,$$

equivalently $d_A^ F_A = 0$. Relate the abelian case $G = U(1)$ to the source-free Maxwell equation from the Hodge theory subsection.*

The Standard Model is specified here by the gauge group

$$G_{\text{SM}} = \text{SU}(3)_C \times \text{SU}(2)_L \times \text{U}(1)_Y.$$

Matter fields are sections of associated bundles for representations of G_{SM} . Write $(\mathbf{n}, \mathbf{m})_Y$ for the representation of dimension n under $\text{SU}(3)_C$, dimension m under $\text{SU}(2)_L$, and weak hypercharge Y . One generation of Weyl fermion multiplets is

$$\begin{aligned} Q_L &= (\mathbf{3}, \mathbf{2})_{1/6}, & u_R &= (\mathbf{3}, \mathbf{1})_{2/3}, & d_R &= (\mathbf{3}, \mathbf{1})_{-1/3}, \\ L_L &= (\mathbf{1}, \mathbf{2})_{-1/2}, & e_R &= (\mathbf{1}, \mathbf{1})_{-1}. \end{aligned}$$

Define the electric-charge operator by $Q = T_3 + Y$.

Let ϕ be a section of the Higgs bundle associated to $(\mathbf{1}, \mathbf{2})_{1/2}$, with potential

$$V(\phi) = -\mu^2 |\phi|^2 + \lambda |\phi|^4$$

for $\mu^2, \lambda > 0$. Write g and g' for the $\text{SU}(2)_L$ and $\text{U}(1)_Y$ coupling constants, and let

$$D\phi = d\phi + \frac{i}{2}g W^a \sigma_a \phi + \frac{i}{2}g' B\phi$$

be the electroweak covariant derivative.

Exercise 58 (Standard Model representations and symmetry breaking)

Determine the Standard Model Lie algebra and decompose a connection into its gluon, weak-boson, and hypercharge components. Verify the electric charges of the fermions above. Show that a Higgs vacuum proportional to $(0, 1)$ has infinitesimal stabilizer $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\text{EM}}$ and breaks the electroweak group by

$$\text{SU}(2)_L \times \text{U}(1)_Y \longrightarrow \text{U}(1)_{\text{EM}}.$$

Derive the W and Z masses, prove that the photon remains massless, and prove cancellation of $\text{Tr}(Y)$ and $\text{Tr}(Y^3)$ for one fermion generation, treating right-handed fields as left-handed conjugates.

2.7 Chern–Simons theory and TQFT

For two connections on E , set

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \quad A = \nabla^1 - \nabla^0, \quad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \text{End } E)$, and define

$$\text{CS}_P(\nabla^0, \nabla^1) = k \int_0^1 P(A, F_t, \dots, F_t) dt.$$

This defines $\text{CS}_P(\nabla^0, \nabla^1) \in \Omega^{2k-1}(M)$.

Exercise 59 (Chern–Simons transgression)

Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d \text{CS}_P(\nabla^0, \nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M, \mathfrak{g})$, wedge products include matrix multiplication. As above, write

$$F_A = dA + A \wedge A$$

and define

$$\text{CS}(A) = \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

Exercise 60 (Chern–Simons form)

Prove that

$$d \text{CS}(A) = \text{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M , define

$$S_{\text{CS}}(A) = \int_M \text{CS}(A).$$

Exercise 61 (Chern–Simons equation)

For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left. \frac{d}{dt} \right|_{t=0} S_{\text{CS}}(A_t) = 2 \int_M \text{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \rightarrow G$, define

$$A^g = g^{-1} dg + g^{-1} dg.$$

Assume that G is compact, connected, simple, and simply connected, and normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2} \int_M \text{tr}((g^{-1} dg)^3) \in \mathbb{Z}.$$

For $k \in \mathbb{R}$, set $S_k = (k/4\pi) S_{\text{CS}}$.

Exercise 62 (Gauge transformation)

Prove that

$$\text{CS}(A^g) - \text{CS}(A) = -d \, \text{tr}(dg \, g^{-1} \wedge A) - \frac{1}{3} \text{tr}((g^{-1}dg)^3).$$

Deduce that $\exp(iS_k)$ is invariant under all gauge transformations when $k \in \mathbb{Z}$.

Now restrict to $2 + 1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2, \mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2, \mathbb{R})), \quad A^\pm = \omega \pm \frac{1}{\ell} e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \quad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 63 (Flat connections encode three-dimensional gravity)

Prove that $F_{A^+} = F_{A^-} = 0$ is equivalent to

$$d_\omega e = 0, \quad F_\omega + \frac{1}{\ell^2} e \wedge e = 0.$$

Assuming that $e : TM \rightarrow \mathfrak{sl}(2, \mathbb{R})$ is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.

Define the first-order three-dimensional Einstein–Hilbert functional, up to its conventional overall coefficient, by

$$S_{\text{EH}}(e, \omega) = \int_M \text{tr} \left(e \wedge F_\omega + \frac{1}{3\ell^2} e \wedge e \wedge e \right).$$

Exercise 64 (Three-dimensional gravity is a Chern–Simons theory)

Prove that, up to an overall normalization and an exact term,

$$S_{\text{CS}}(A^+) - S_{\text{CS}}(A^-)$$

equals $S_{\text{EH}}(e, \omega)$ [19, 20]. Explain why this is a local Lie-algebra formulation in three spacetime dimensions and why changing between Lorentzian and Euclidean signatures changes the real form and reality conditions rather than deriving four-dimensional gravity by Wick rotation.

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\text{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \text{tr}_V(\text{Hol}_\gamma(A)).$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \dots, V_r$, set

$$W_L(A) = \prod_{j=1}^r W_{V_j}(\gamma_j, A).$$

For a framed link, define the formal Chern–Simons expectation value [17]

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A) \mathcal{D}A}{\int e^{iS_k(A)} \mathcal{D}A}$$

For normalized $SU(2)$ theory at integer level $k \geq 1$, set

$$q = \exp\left(\frac{2\pi i}{k+2}\right).$$

Exercise 65 (Wilson loops and the Jones polynomial)

Prove that the expectation value depends on the chosen framing. Let L_+ , L_- , and L_0 differ only by a positive crossing, a negative crossing, and the oriented smoothing. Assume that, after the standard framing normalization, the Wilson-loop expectation values satisfy

$$q^{-1}\langle W_{L_+} \rangle_k - q\langle W_{L_-} \rangle_k = \left(q^{1/2} - q^{-1/2}\right) \langle W_{L_0} \rangle_k$$

and $\langle W_{\bigcirc} \rangle_k = 1$. Prove that these relations determine the Jones polynomial of an oriented link.

Let $\text{Bord}_3^{\text{or}}$ be the symmetric monoidal category whose objects are closed oriented surfaces, whose morphisms are oriented three-dimensional bordisms modulo diffeomorphism, and whose tensor product is disjoint union. A **three-dimensional oriented TQFT** is a symmetric monoidal functor

$$Z : \text{Bord}_3^{\text{or}} \longrightarrow \text{Vect}_{\mathbb{C}}.$$

Exercise 66 (Reshetikhin–Turaev TQFT)

Let $\mathcal{C}$ be a modular braided fusion category. Prove that the Reshetikhin–Turaev construction assigns finite-dimensional state spaces to closed oriented surfaces and compatible linear maps to bordisms, defining a three-dimensional oriented TQFT [18]. Prove that labeled Wilson lines induce the braid-group actions determined by the F - and R -symbols of $\mathcal{C}$. For the $SU(2)_k$ category, relate the resulting link invariant to the Chern–Simons Wilson-loop skein relation above.

3 Analysis

3.1 Norms, operator exponentials, and linear evolution

A **norm** on a complex vector space H is a function $\|\cdot\| : H \rightarrow [0, \infty)$ that is definite, homogeneous, and satisfies the triangle inequality. A normed space

is a **Banach space** if every Cauchy sequence converges. A **Hilbert space** is a complex vector space with a Hermitian inner product $\langle \cdot, \cdot \rangle$ whose norm $\|x\| = \sqrt{\langle x, x \rangle}$ is complete. A linear map $A : H \rightarrow K$ is **bounded** if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty.$$

Write $\mathcal{B}(H, K)$ for the normed space of bounded operators and $\mathcal{B}(H) = \mathcal{B}(H, H)$.

For $A \in \mathcal{B}(H)$, define the operator exponential by the formal series

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

whenever the series converges in operator norm.

Exercise 67 (Cauchy–Schwarz inequality)

Prove that every inner-product space satisfies

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Deduce the triangle inequality for the induced norm. Prove that $\mathcal{B}(H, K)$ is complete when K is complete and that the exponential series converges in $\mathcal{B}(H)$ for every bounded A .

Exercise 68 (Operator exponentials and linear evolution)

Prove that e^{tA} is differentiable in operator norm and that

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

Deduce that the unique solution of

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0,$$

is $x(t) = e^{tA} x_0$. In finite dimensions use the Jordan decomposition

$$A = S + N, \quad [S, N] = 0,$$

with S semisimple and N nilpotent, to prove

$$e^{tA} = e^{tS} e^{tN}.$$

Show that on the generalized λ -eigenspace every component of a solution is a linear combination of terms $t^j e^{\lambda t}$.

3.2 Hilbert-space operator theory

An **adjoint** of $A \in \mathcal{B}(H, K)$ is an operator $A^* \in \mathcal{B}(K, H)$ satisfying

$$\langle Ax, y \rangle = \langle x, A^* y \rangle.$$

An operator $A \in \mathcal{B}(H)$ is **self-adjoint** if $A = A^*$, **normal** if $A^* A = A A^*$, **positive** if $\langle x, Ax \rangle \geq 0$ for every x , and **unitary** if $A^* A = A A^* = 1$.

Exercise 69 (Adjoint of bounded operators)

Prove that every bounded operator between Hilbert spaces has a unique adjoint and establish the identities

$$(AB)^* = B^*A^*, \quad \|A^*A\| = \|A\|^2.$$

3.3 Compact operators and the Fredholm alternative

A bounded operator $K : H \rightarrow L$ between Hilbert spaces is **compact** if the image of the closed unit ball has compact closure. Write $\mathcal{K}(H, L)$ for the compact operators and $\mathcal{K}(H) = \mathcal{K}(H, H)$.

Exercise 70 (Compact operators)

Prove that K is compact if and only if, for every bounded sequence (x_n) in H , the sequence (Kx_n) has a convergent subsequence. Prove that every finite-rank operator is compact, that $\mathcal{K}(H, L)$ is closed in the operator norm, and that

$$A \in \mathcal{B}(L, M), \quad K \in \mathcal{K}(H, L), \quad B \in \mathcal{B}(N, H) \implies AKB \in \mathcal{K}(N, M).$$

Deduce that $\mathcal{K}(H)$ is a closed two-sided ideal in $\mathcal{B}(H)$. Prove that K^* is compact whenever K is compact and that, on a Hilbert space, every compact operator is an operator-norm limit of finite-rank operators.

A bounded operator $T : H \rightarrow L$ is **Fredholm** if $\ker T$ and $L/\operatorname{im} T$ are finite dimensional and $\operatorname{im} T$ is closed. Its **Fredholm index** is

$$\operatorname{ind}(T) = \dim \ker T - \dim(L/\operatorname{im} T).$$

Exercise 71 (Fredholm alternative)

Let $K \in \mathcal{K}(H)$ and $\lambda \in \mathbb{C} \setminus \{0\}$. Prove that $\lambda 1 - K$ has closed range and that [2]

$$\dim \ker(\lambda 1 - K) = \dim \ker(\bar{\lambda} 1 - K^*) < \infty.$$

Using

$$\operatorname{im}(\lambda 1 - K)^\perp = \ker(\bar{\lambda} 1 - K^*),$$

prove that the following are equivalent:

- $\lambda 1 - K$ is injective.
- $\lambda 1 - K$ is surjective.
- $\lambda 1 - K$ is invertible.

Prove more generally that $(\lambda 1 - K)x = y$ is solvable if and only if y is orthogonal to $\ker(\bar{\lambda} 1 - K^*)$. Deduce that $\lambda 1 - K$ is Fredholm of index zero. Finally, prove that every nonzero point of the spectrum of a compact operator is an eigenvalue of finite multiplicity and that zero is the only possible accumulation point of the spectrum.

3.4 Quantum states and uncertainty

A finite-dimensional **quantum state** is a positive operator ρ with $\text{tr}(\rho) = 1$. A unit vector ψ defines the pure state $\rho_\psi = |\psi\rangle\langle\psi|$. For a self-adjoint observable A , define

$$\langle A \rangle_\rho = \text{tr}(\rho A), \quad \text{Var}_\rho(A) = \text{tr}(\rho(A - \langle A \rangle_\rho 1)^2).$$

For $\rho = \rho_\psi$, use the subscripts ψ instead of ρ .

Exercise 72 (Robertson uncertainty inequality)

For self-adjoint $A, B \in \mathcal{B}(H)$ and a unit vector ψ , prove [12]

$$\text{Var}_\psi(A) \text{Var}_\psi(B) \geq \frac{1}{4} |\langle \psi, [A, B] \psi \rangle|^2, \quad [A, B] = AB - BA.$$

Determine the equality condition.

For a self-adjoint H , write $U_t = e^{-itH}$.

Exercise 73 (Commuting observables are conserved)

Let A, H be self-adjoint. Prove that U_t is unitary and that

$$\frac{d}{dt} \langle U_t \psi, A U_t \psi \rangle = -i \langle U_t \psi, [H, A] U_t \psi \rangle.$$

Deduce that $[H, A] = 0$ if and only if the expectation of A is constant under this evolution for every initial vector.

Exercise 74 (No-cloning theorem)

Let $b \in H$ be a unit vector and suppose that a unitary $C : H \otimes H \rightarrow H \otimes H$ satisfies $C(v \otimes b) = v \otimes v$ and $C(w \otimes b) = w \otimes w$ for unit vectors v, w . Prove that [11]

$$\langle v, w \rangle = \langle v, w \rangle^2.$$

Deduce that no unitary map clones every unit vector.

3.5 Schmidt decomposition and entanglement

For finite-dimensional Hilbert spaces H_A, H_B and $\psi \in H_A \otimes H_B$, define

$$T_\psi : H_A^* \rightarrow H_B, \quad T_\psi(\phi) = (\phi \otimes 1)(\psi).$$

A **Schmidt decomposition** of ψ is an expression

$$\psi = \sum_{i=1}^r s_i a_i \otimes b_i.$$

where (a_i) and (b_i) are orthonormal families and $s_i > 0$; r is the **Schmidt rank**. For a unit vector ψ , define the reduced density operators by

$$\rho_A = \text{Tr}_{H_B}(|\psi\rangle\langle\psi|), \quad \rho_B = \text{Tr}_{H_A}(|\psi\rangle\langle\psi|).$$

Exercise 75 (Schmidt decomposition and entanglement)

Use the singular-value decomposition of T_ψ to prove that every ψ has a Schmidt decomposition. Prove that the nonzero eigenvalues of ρ_A and ρ_B are the numbers s_i^2 , with the same multiplicities. Prove that ψ is separable, meaning $\psi = a \otimes b$, exactly when its Schmidt rank is one. Deduce that the entanglement entropy is

$$S(\psi) = -\text{Tr}(\rho_A \log \rho_A) = -\sum_i s_i^2 \log s_i^2,$$

with the convention $0 \log 0 = 0$.

3.6 CHSH, Gisin, and Tsirelson

Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$. Their **CHSH operator** is

$$\mathcal{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Exercise 76 (Every entangled pure two-qubit state violates CHSH)

Prove that every classical local-hidden-variable assignment satisfies $|\langle \mathcal{C} \rangle| \leq 2$. For the two-qubit state in Schmidt form

$$\psi_\theta = \cos \theta |00\rangle + \sin \theta |11\rangle, \quad 0 \leq \theta \leq \frac{\pi}{4},$$

optimize over projective binary observables and prove that the maximal CHSH value is $[\gamma]$

$$2\sqrt{1 + \sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates a CHSH inequality. For $\theta = \pi/4$, recover the value $2\sqrt{2}$.

A **bipartite binary quantum correlation matrix** has entries

$$c_{ij} = \langle \psi, (A_i \otimes B_j) \psi \rangle,$$

where ψ is a unit vector. The operators A_i and B_j are self-adjoint unitaries on finite-dimensional Hilbert spaces.

For a real matrix $M = (M_{ij})$, define

$$\beta_C(M) = \max_{\varepsilon_i, \delta_j \in \{-1, 1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_Q(M) = \sup_c \left| \sum_{i,j} M_{ij} c_{ij} \right|,$$

where the supremum is over bipartite binary quantum correlation matrices.

Exercise 77 (Tsirelson's representation theorem and bound)

Prove that a finite real matrix $c = (c_{ij})$ is a bipartite binary quantum correlation matrix if and only if there are unit vectors x_i, y_j in a real Hilbert space such that [8]

$$c_{ij} = \langle x_i, y_j \rangle.$$

Deduce that

$$\beta_Q(M) = \sup_{\|x_i\|=\|y_j\|=1} \left| \sum_{i,j} M_{ij} \langle x_i, y_j \rangle \right|.$$

For the CHSH operator, prove the Tsirelson bound $\|C\| \leq 2\sqrt{2}$ and exhibit observables and a state attaining equality.

3.7 Grothendieck inequality

The **real Grothendieck constant** $K_G^{\mathbb{R}}$ is the least constant for which

$$\beta_Q(M) \leq K_G^{\mathbb{R}} \beta_C(M)$$

holds for every real matrix M .

Exercise 78 (Grothendieck's inequality)

Prove that $K_G^{\mathbb{R}} < \infty$ [9, 10]. For

$$M_{\text{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

compute $\beta_C(M_{\text{CHSH}})$ directly. Choose four unit vectors in $\mathbb{R}^2$ that attain

$$\beta_Q(M_{\text{CHSH}}) = 2\sqrt{2}$$

and verify the ratio $\sqrt{2}$. Deduce that $\sqrt{2} \leq K_G^{\mathbb{R}}$ and interpret the inequality as a dimension-independent bound on quantum advantage for bipartite binary correlation Bell inequalities.

3.8 Von Neumann algebras and subfactors

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$ -subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \rightarrow T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta \rangle \longrightarrow \langle T\xi, \eta \rangle$$

for every $\xi, \eta \in H$.

Exercise 79 (Bicommutant theorem)

Let $A \subseteq \mathcal{B}(H)$ be a unital $*$ -subalgebra. Prove that

$$A'' = \overline{A}^{\text{SOT}} = \overline{A}^{\text{WOT}}.$$

The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$. A positive functional or positive map is **normal** if it preserves suprema of bounded increasing nets of positive elements, and a positive functional ψ is **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type II_1 factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \longrightarrow \mathbb{C}, \quad \tau(ab) = \tau(ba), \quad \tau(1) = 1.$$

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \quad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 80 (Projection dimension)

Let M be a type II_1 factor with normalized trace τ . Prove that

$$p \sim q \implies \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \iff \tau(p) \leq \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. For a linear map $\Phi : M \rightarrow N$, define its matrix amplification by $\Phi_n([x_{ij}]) = [\Phi(x_{ij})]$; the map is **completely positive** if every Φ_n is positive. Let $N \subseteq M$ be finite von Neumann algebras with a common normalized trace τ . A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \rightarrow N$ such that

$$\tau \circ E_N = \tau, \quad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^*x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 81 (Conditional expectation and Jones projection)

Prove that there is a unique trace-preserving conditional expectation $E_N : M \rightarrow N$. Prove that E_N extends to the orthogonal projection

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \quad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N . For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N(pL^2(N, \tau)^k) = \sum_{i=1}^k \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M, \tau)$ is isomorphic as a right N -module to such a module; then

$$[M : N] = \dim_N L^2(M, \tau).$$

Exercise 82 (Jones basic construction)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [3]

$$\tau_1(xe_N y) = \frac{1}{[M : N]} \tau(xy)$$

for $x, y \in M$.

The **Jones tower** is the sequence obtained by iterating the basic construction:

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots.$$

Let e_i be the Jones projection used to construct M_i from M_{i-1} , and set

$$\delta = \sqrt{[M : N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \quad U_i U_j = U_j U_i \quad (|i - j| \geq 2), \quad U_i U_{i \pm 1} U_i = U_i.$$

Exercise 83 (Temperley–Lieb relations)

Prove that the Jones projections satisfy [6, 4]

$$e_i^2 = e_i = e_i^*, \quad e_i e_j = e_j e_i \quad (|i - j| \geq 2),$$

$$e_i e_{i \pm 1} e_i = \delta^{-2} e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define

$$\sigma_i = q 1 - U_i.$$

Exercise 84 (Braid representation)

Prove that σ_i is invertible with inverse $q^{-1}1 - U_i$, and prove that

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1},$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [5].

A **Markov trace** on the Temperley–Lieb tower is a compatible family of positive normalized traces satisfying

$$\mathrm{tr}(xU_i) = \delta^{-1} \mathrm{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators. Define

$$P_0(\delta) = 1, \quad P_1(\delta) = \delta, \quad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Exercise 85 (Positivity quantizes subfactor indices below four)

Prove that the normalized traces on the Jones tower restrict to a Markov trace. Prove that, when $\sin \theta \neq 0$,

$$P_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$, then

$$\delta = 2 \cos \left(\frac{\pi}{m} \right)$$

for some integer $m \geq 3$.

Exercise 86 (Jones index theorem)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that [4]

$$[M : N] \in \left\{ 4 \cos^2 \left(\frac{\pi}{n} \right) : n \geq 3 \right\} \cup [4, \infty).$$

For

$$[M : N] = 4 \cos^2 \left(\frac{\pi}{k+2} \right), \quad \delta = \sqrt{[M : N]},$$

prove that the Jones projections determine a representation of the Temperley–Lieb algebra $\mathrm{TL}_n(\delta)$, and hence a braid-group representation. Show that the Markov trace of a braid closure satisfies the Jones skein relation. Relate the parameter

$$\delta = 2 \cos \left(\frac{\pi}{k+2} \right)$$

to the $SU(2)_k$ fusion and braiding data from Algebra and to the Wilson-loop invariant of Chern–Simons theory from Geometry.

References

- [1] Alexei Kitaev, *Periodic table for topological insulators and superconductors*, AIP Conference Proceedings, 1134:22–30, 2009. doi:10.1063/1.3149495.
- [2] John B. Conway, *A Course in Functional Analysis*, 2nd ed., Graduate Texts in Mathematics, Vol. 96, Springer, 1990.
- [3] Vaughan F. R. Jones and V. S. Sunder, *Introduction to Subfactors*, London Mathematical Society Lecture Note Series, Vol. 234, Cambridge University Press, 1997. doi:10.1017/CBO9780511566219.
- [4] Vaughan F. R. Jones, *Index for subfactors*, Inventiones Mathematicae, 72(1):1–25, 1983. doi:10.1007/BF01389127.
- [5] Vaughan F. R. Jones, *A polynomial invariant for knots via von Neumann algebras*, Bulletin of the American Mathematical Society, 12(1):103–111, 1985. doi:10.1090/S0273-0979-1985-15304-2.
- [6] H. N. V. Temperley and E. H. Lieb, *Relations between the “percolation” and “colouring” problem and other graph-theoretical problems associated with regular planar lattices*, Proceedings of the Royal Society A, 322(1549):251–280, 1971. doi:10.1098/rspa.1971.0067.
- [7] Nicolas Gisin, *Bell’s inequality holds for all non-product states*, Physics Letters A, 154(5–6):201–202, 1991. doi:10.1016/0375-9601(91)90805-I.
- [8] B. S. Tsirelson, *Quantum generalizations of Bell’s inequality*, Letters in Mathematical Physics, 4:93–100, 1980. doi:10.1007/BF00417500.
- [9] Alexander Grothendieck, *Résumé de la théorie métrique des produits tensoriels topologiques*, Boletim da Sociedade de Matemática de São Paulo, 8:1–79, 1953.
- [10] Gilles Pisier, *Grothendieck’s theorem, past and present*, Bulletin of the American Mathematical Society, 49(2):237–323, 2012. doi:10.1090/S0273-0979-2011-01348-9.
- [11] W. K. Wootters and W. H. Zurek, *A Single Quantum Cannot Be Cloned*, Nature, 299:802–803, 1982. doi:10.1038/299802a0.
- [12] H. P. Robertson, *The Uncertainty Principle*, Physical Review, 34:163–164, 1929. doi:10.1103/PhysRev.34.163.
- [13] Emmy Noether, *Invariant Variation Problems*, translated by M. A. Tavel, Transport Theory and Statistical Physics, 1(3):186–207, 1971; originally published 1918. doi:10.1080/00411457108231446.
- [14] Manfredo P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976.

- [15] S. S. Chern and J. Simons, *Characteristic forms and geometric invariants*, Annals of Mathematics, 99(1):48–69, 1974. doi:10.2307/1971013.
- [16] C. N. Yang and R. L. Mills, *Conservation of Isotopic Spin and Isotopic Gauge Invariance*, Physical Review, 96(1):191–195, 1954. doi:10.1103/PhysRev.96.191.
- [17] E. Witten, *Quantum field theory and the Jones polynomial*, Commun. Math. Phys., 121(3):351–399, 1989. doi:10.1007/BF01217730.
- [18] Nicolai Reshetikhin and Vladimir G. Turaev, *Ribbon graphs and their invariants derived from quantum groups*, Communications in Mathematical Physics, 127(1):1–26, 1990. doi:10.1007/BF02096491.
- [19] A. Achúcarro and P. K. Townsend, *A Chern–Simons action for three-dimensional anti-de Sitter supergravity theories*, Physics Letters B, 180(1–2):89–92, 1986. doi:10.1016/0370-2693(86)90140-1.
- [20] Edward Witten, *2 + 1 dimensional gravity as an exactly soluble system*, Nuclear Physics B, 311(1):46–78, 1988. doi:10.1016/0550-3213(88)90143-5.
- [21] Yakir Aharonov and David Bohm, *Significance of Electromagnetic Potentials in the Quantum Theory*, Physical Review, 115(3):485–491, 1959. doi:10.1103/PhysRev.115.485.
- [22] Parsa H. Bonderson, *Non-Abelian Anyons and Interferometry*, Ph.D. thesis, California Institute of Technology, 2007. doi:10.7907/5NDZ-W890.
- [23] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, *Direct observation of anyonic braiding statistics*, Nature Physics, 16:931–936, 2020. doi:10.1038/s41567-020-1019-1.
- [24] M. Iqbal et al., *Non-Abelian topological order and anyons on a trapped-ion processor*, Nature, 626:505–511, 2024. doi:10.1038/s41586-023-06934-4.
- [25] S. Xu et al., *Non-Abelian braiding of Fibonacci anyons with a superconducting processor*, Nature Physics, 20:1469–1475, 2024. doi:10.1038/s41567-024-02529-6.